TOP-K DOCUMENT RECOMMENDER

Organizational Reflection

Team members

Frida Garza
Pedro de Leon
Paola Urdiales

Instructor

Jesus Chio

Python · TF-IDF · Cosine similarity

What did we build?

Python · 31 simulated documents · TF-IDF and cosine similarity

The user provides a document ID or free text to receive the Top-5 matches.

Examples from the project README

README_usage_examples

```
recommend(seed_id="DOC014")
recommend(query="I can't log in because I forgot my password")
recommend(seed_id="DOC014", k=3)
```

Source: GitHub README.md · Actual command examples (not a notebook screenshot)

Why does it help organizations?

Less rework: reuse existing guidance. Faster decisions: retrieve related documents immediately.

Real Top-5 output for VPN incident DOC014

README_actual_top5_result

INPUT (ID): DOC014

#	ID	Score	Category	Title
1	DOC009	0.3236	Policy/Process	Remote Work and VPN Access Policy
2	DOC026	0.1445	FAQ	FAQ: How do I connect to the network from home?
3	DOC013	0.1214	Support/Incidents	Ticket: User Locked Out Due to Expired Password
4	DOC020	0.1171	Support/Incidents	Incident: Network Connectivity Loss at the North Branch
5	DOC016	0.0964	Support/Incidents	Incident: Phishing Attempt Reported by Employees

Most relevant: DOC009, Remote Work and VPN Access Policy · cosine similarity 0.3236

What are the limitations?

Synonyms

TF-IDF may miss related ideas expressed with different words.

Ambiguity

Shared keywords do not always mean the same topic or intent.

Data quality

Short texts and a limited simulated corpus can weaken results.

What would we improve?

Metadata and filters

Filter by category, date, team, or relevance.

Access control

Only show documents a user is authorized to view.

Better semantic search

Test embeddings to recognize similar meanings.

Continuous evaluation

Add new documents and monitor recommendation quality.

Bottom line: a useful prototype, with room to improve for real organizational use.